

QUERYING TABLES

The SELECT command retrieves data from tables based on the query requests

5	SELECT [DISTINCT] (* / column1 [alias], expressions [alias])
1	FROM table1 [alias], table2, ...
2	WHERE (conditions, specifying the connections between the tables)
3	GROUP BY table_name.column_name
4	HAVING (conditions imposed on group values)
6	ORDER BY table_name.column_name ASC/DESC;

where:

SELECT	specifies the selected attributes;
DISTINCT	suppresses duplicate values;
*	selects all attributes;
column	selects a specific column or columns in one or more tables;
expression	allows the construction of expressions resulting in new values
alias	alternative names for selected columns;
FROM tables	specifies the tables that contain the selected columns;;
WHERE	the clause allows specifying the conditions and criteria for data selection;
GROUP BY	specifies the field according to which the data will be grouped in the case of group expressions and functions (SUM(), AVG(), COUNT(), MIN(), MAX());
HAVING	in the case of group functions, the conditions imposed on them are specified in the HAVING clause;
ORDER BY	specifies the sorting according to a certain field ascending (ASC) – default or descending (DESC). Only ORDER BY allows the use of the alias;

SQL commands:

- Are not case sensitive, usually the commands are written in capital letters to make it easier to distinguish them;
- Can be written on several lines;
- Keywords cannot be abbreviated or written on several lines;

The following operators are used in the construction of SQL commands:

<, >, =, >=, <=, NOT	Comparison operators
BETWEEN val_min AND val_max	Between 2 values.
SELECT employee_id, last_name, salary, department_id FROM employees WHERE salary BETWEEN 2000 AND 6100 AND department_id NOT IN (20, 30); SELECT employee_id, last_name, salary, department_id FROM employees WHERE salary >= 2000 AND salary <=6100 AND department_id = 50;	
IN (list of values)	equal to any value in the list of values
SELECT employee_id, last_name, manager_id FROM employees WHERE manager_id	

IN (120, 122, 123);	
LIKE	similar to a template % any characters; _ one character;
SELECT employee_id, last_name, job_id FROM employees WHERE last_name LIKE 'B%'; SELECT employee_id, last_name, job_id FROM employees WHERE last_name LIKE '__t%';	
IS NULL	has the value NULL
SELECT employee_id, last_name, commission_pct FROM employees WHERE commission_pct IS NULL;	

The columns are specified in the order in which they are to be displayed, not necessarily in the order in which they appear in the description of the tables.

The value NULL, as in other languages, means unavailable, unallocated value, and is not the same as "blank" or "zero".

With the help of the SELECT command, information can be extracted from the database. The SELECT command allows the implementation of relational algebra operators: selection, projection, join, union, intersection, difference.

Selection: Allows you to obtain a new relationship that contains all the attributes of the initial relationship and a smaller number of tuples. The reduction is made according to a condition called the selection condition.

SELECT * FROM original_table
WHERE condition;

Projection: Allows you to obtain a new relationship that contains a smaller number of attributes compared to the initial relationship and all distinct values or combinations of values of these attributes.

SELECT [DISTINCT] column_list
FROM original_table;

Join: Binary operator, a particular case being the Cartesian product. The resulting relationship will contain the tuples corresponding to the join attributes that are in a given relationship. The most commonly used join case is one in which join attributes are the primary key in one relationship and the foreign key in the other relationship..

Exercises

1. Display all the employees from the Employees table

`SELECT * FROM employees;`

2. Display the employee_id, last_name, first_name and job id from Employees.

`SELECT employee_id, last_name, first_name, job_id FROM employees;`

3. Display only the employees who are part of the civil servant category (whose job id contains "CLERK").
4. Display the orders concluded by the employee with employee_id = 161
5. Display all orders that were launched online after January 1, 2000.
6. Select employee_id, last name, first name, manager_id, department_id from the employees table and department_name from the departments table and join the two tables.

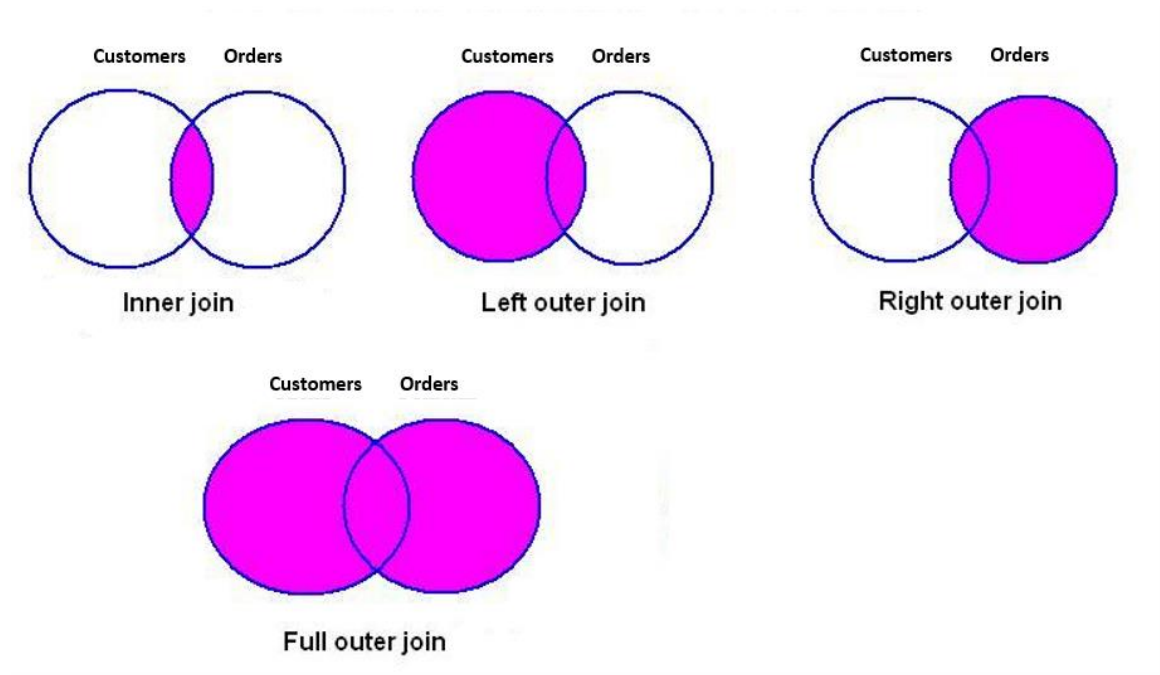
Query Result x

SQL | Fetched 50 rows in 0.02 seconds

	EMPLOYEE_ID	LAST_NAME	FIRST_NAME	MANAGER_ID	DEPARTMENT_ID	DEPARTMENT_NAME
1	200	Whalen	Jennifer	101	10	Administration
2	201	Hartstein	Michael	100	20	Marketing
3	202	Fay	Pat	201	20	Marketing
4	114	Raphaely	Den	100	30	Purchasing
5	115	Khoo	Alexander	114	30	Purchasing
6	116	Baida	Shelli	114	30	Purchasing
7	117	Tobias	Sigal	114	30	Purchasing
8	118	Himuro	Guy	114	30	Purchasing
9	119	Colmenares	Karen	114	30	Purchasing
10	203	Mavris	Susan	101	40	Human Resources
11	120	Weiss	Matthew	100	50	Shipping

Join Types:

- ❑ **INNER JOIN**
- ❑ **OUTER JOIN**
 - **Left**
 - **Right**
 - **Full**



EQUI JOIN

Join condition:

table1.join_attribute=table2.join_attribute

OUTER JOIN

Left outer join

table1.join_attribute=table2.join_attribute(+)

Right outer join

table1.join_attribute(+)=table2.join_attribute

Full outer join

Left outer join command

UNION

Right outer join command

1. Display the orders concluded by your employees (in the WHERE clause you will specify the join condition between the tables).

```
SELECT e.*, o.*
FROM employees e, orders o
WHERE e.employee_id= o.sales_rep_id;
```

2. Display the orders managed by the employee named Greene only in the month of November (last_name).
3. Calculate the value of each product and display the product name, price quantity and value.
4. Display only the products with a values between 1000 and 2000.
5. Display the product id, product name and quantity for all products, even if they have not been ordered.

Subqueries (using 2 nested SELECT commands)

6. Display the products that have the lowest unit price:

```
SELECT p.product_name, oi.unit_price
FROM product_information p, order_items oi
WHERE p.product_id = oi.product_id
AND oi.unit_price=(SELECT MIN(unit_price) FROM order_items);
```

7. Display the employees working in the same department as employee Smith (last_name).

The ANY and the ALL operators

ANY compares the value with any value returned by the query
 ALL compares the value with each value returned by the query

8. Display the employee_id, first name, job_id and salary for employees who do not work as IT_PROG and whose salary is lower than any of the salaries of employees working as IT_PROG:

```
SELECT employee_id, first_name, job_id, salary
FROM employees
Where salary < ANY(Select salary FROM employees where
                    job_id='IT_PROG')
AND job_id<>'IT_PROG'
ORDER BY salary DESC;
```

	SALARY
1	9000
2	6000
3	4800
4	4800
5	4200

9. Display the employee_id, first name, job_id and salary for employees who do not work as IT_PROG and whose salary is lower than each of the salaries of employees working as IT_PROG:

```
SELECT employee_id, first_name, job_id, salary
FROM employees
Where salary < ALL(Select salary FROM employees where
                   job_id='IT_PROG')
AND job_id<>'IT_PROG' ORDER BY salary DESC;
```